Economics S5040: Dynamic Stochastic Modeling
Problem Set 7 — Optimal Saving

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Instructions

Submit one Jupyter notebook containing all derivations, code, figures, numerical answers, and economic explanations.

This problem set has three substantive problems.

1. Cake eating: solve a simple optimal saving problem analytically and numerically.
2. Stochastic neoclassical growth: calibrate and solve a dynamic program under uncertainty.
3. Saving with labor income: solve a small finite-state consumption-saving problem and interpret the policy.

Problem 1. Cake Eating: Analytical and Numerical Solution

A household begins with a stock $x_0 > 0$ of a perfectly durable good. It chooses consumption $\{c_t\}_{t=0}^{\infty}$ to solve

$$\max_{\{c_t\}} \sum_{t=0}^{\infty} \beta^t u(c_t), \qquad 0 < \beta < 1,$$

subject to

$$x_{t+1} = x_t - c_t, \qquad 0 \le c_t \le x_t.$$

Use CRRA utility

$$u(c) = \begin{cases} \dfrac{c^{1-\rho} - 1}{1-\rho}, & \rho \neq 1, \\ \log c, & \rho = 1. \end{cases}$$

(a) Write the Bellman equation.

(b) Derive the Euler equation

$$u'(c_t) = \beta u'(c_{t+1}).$$

(c) Show that with CRRA utility,

$$\frac{c_{t+1}}{c_t} = \beta^{1/\rho}.$$

(d) Show that the optimal policy is

$$\boxed{c^*(x) = \left(1 - \beta^{1/\rho}\right) x}$$

and hence the saving policy is

$$\boxed{a^*(x) = \beta^{1/\rho} x.}$$

(e) Let

$$\beta = 0.95, \qquad \rho = 2.$$

Construct a grid for x on $[10^{-4}, 10]$. Solve the Bellman equation numerically by value function iteration using a continuous consumption choice and linear interpolation of the value function.

Plot:

- the numerical consumption policy;
- the exact consumption policy;
- the absolute policy error.

Report the maximum absolute policy error.

(f) Repeat the numerical solution with $\beta = 0.90$ and $\beta = 0.98$. Explain how patience changes the consumption and saving policies.

Problem 2. Stochastic Neoclassical Growth

Consider the neoclassical growth model in quarterly time.

Production:

$$y_t = A z_t k_t^{\alpha}.$$

Resource constraint:

$$c_t + k_{t+1} = A z_t k_t^{\alpha} + (1 - \delta) k_t.$$

Preferences:

$$\max_{\{c_t, k_{t+1}\}} E_0 \sum_{t=0}^{\infty} \beta^t \log(c_t).$$

Throughout, use

$$\alpha = 0.33, \qquad \beta = 0.96, \qquad \delta = 0.10, \qquad k_{ss} = 400,$$

where capital is measured in thousands of 2017 dollars per worker.

2.1 Deterministic Model and Calibration

First set $z_t \equiv 1$.

(a) Derive the Euler equation. Show that

$$\frac{1}{c_t} = \beta \frac{1}{c_{t+1}} \left[\alpha A k_{t+1}^{\alpha - 1} + 1 - \delta\right].$$

(b) In steady state,

$$k_{t+1} = k_t = k^*, \qquad c_{t+1} = c_t = c^*.$$

Show that

$$1 = \beta \left[\alpha A (k^*)^{\alpha - 1} + 1 - \delta \right].$$

Use the calibration target

$$k^* = k_{ss} = 400$$

to solve for

$$\boxed{A = \frac{\frac{1}{\beta} - (1 - \delta)}{\alpha (k^*)^{\alpha - 1}}.}$$

Compute A and steady-state output

$$y^* = A (k^*)^{\alpha}.$$

(c) Write the Bellman equation

$$V(k) = \max_{k' \geq 0} \left\{ \log \left(A k^{\alpha} + (1 - \delta) k - k' \right) + \beta V(k') \right\}.$$

Solve the deterministic dynamic program by value function iteration.

Use a grid for capital centered around k_{ss}.

Plot the policy function

$$k' = g(k)$$

together with the 45-degree line.

Report the tolerance and number of iterations.

Explain why the intersection of the policy function with the 45-degree line identifies the steady state.

2.2 Calibrating the Markov Chain

Now introduce multiplicative productivity uncertainty.

Productivity takes three values:

$$z \in \{0.95, 1.00, 1.026\},$$

corresponding to recession (R), normal (N), and boom (B).

Let P be the 3×3 transition matrix.

(a) If P_{ii} is the probability of remaining in state i, show that the expected duration of a spell in state i is

$$D_i = \frac{1}{1 - P_{ii}}.$$

Use

$$D_R = 4, \qquad D_B = 6$$

to compute P_{RR} and P_{BB}.

(b) Impose

$$P_{RB} = 0.01, \qquad P_{BR} = 0.02, \qquad P_{NR} = 0.03, \qquad P_{NB} = 0.04.$$

Fill in the remaining entries so that each row sums to one.

Verify that

$$P = \begin{pmatrix} 0.75 & 0.24 & 0.01 \\ 0.03 & 0.93 & 0.04 \\ 0.02 & 0.147 & 0.833 \end{pmatrix}.$$

(c) Compute the stationary distribution π satisfying

$$\pi = \pi P, \qquad \sum_i \pi_i = 1.$$

Interpret π economically.

2.3 Solve the Dynamic Program Under Uncertainty

The state is now (k, z).

The Bellman equation is

$$V(k,z) = \max_{k' \geq 0} \left\{ \log\left(Azk^{\alpha} + (1-\delta)k - k'\right) + \beta \sum_{z'} P(z,z')V(k',z') \right\}.$$

(a) Construct a value function array of dimension $3 \times N$ and solve the model by value function iteration.

(b) Compute the policy function $g(k, z)$.

(c) Plot the three policy functions together with the 45-degree line.

Discuss:

- how policy differs across recession, normal, and boom;
- whether higher productivity shifts the policy function upward;
- whether precautionary saving appears in this model.

2.4 Simulation

Simulate the model for

$$T = 100$$

quarters.

Use

$$k_0 = k_{ss}.$$

Draw z_0 from the stationary distribution π.

For $t = 0, \ldots, T-1$:

$$k_{t+1} = g(k_t, z_t),$$

$$y_t = A z_t k_t^{\alpha},$$
$$c_t = A z_t k_t^{\alpha} + (1-\delta)k_t - k_{t+1},$$

and draw z_{t+1} using row z_t of P.

Plot:

- capital k_t;
- consumption c_t;
- shock state z_t.

Discuss how capital responds to recessions and booms, whether the simulated path appears stationary, and how persistence affects the dynamics.

Problem 3. Saving with Labor Income

Consider a household with assets a_t and labor income y_t.

The household chooses next-period assets a_{t+1} according to

$$c_t + a_{t+1} = R a_t + y_t,$$

with

$$a_{t+1} \geq 0.$$

Let

$$R = 1.02, \qquad \beta = 0.96,$$

and use log utility:

$$u(c) = \log c.$$

Income follows a two-state Markov chain:

$$y \in \{1, 2\},$$

with transition matrix

$$P = \begin{pmatrix} 0.90 & 0.10 \\ 0.20 & 0.80 \end{pmatrix}.$$

Use an asset grid from 0 to 10.

(a) Write the Bellman equation

$$V(a, y) = \max_{a' \geq 0} \left\{ \log(Ra + y - a') + \beta E[V(a', y') \mid y] \right\}.$$

State the feasibility restriction on a'.

(b) Solve the problem by value function iteration.

Plot the optimal next-asset policy

$$a' = g(a, y)$$

for both income states.

(c) Plot the consumption policy for both income states.

(d) Compare saving behavior in the low-income and high-income states.

Does the household generally save more when income is high?

Explain why.

(e) Examine the policy near $a = 0$.

Explain the economic meaning of the borrowing constraint

$$a' \geq 0.$$

(f) Briefly compare this model with cake eating.

What new role does saving play once labor income is uncertain?

What surprised me?

Write one short paragraph describing one result from this assignment that surprised you and explain why.